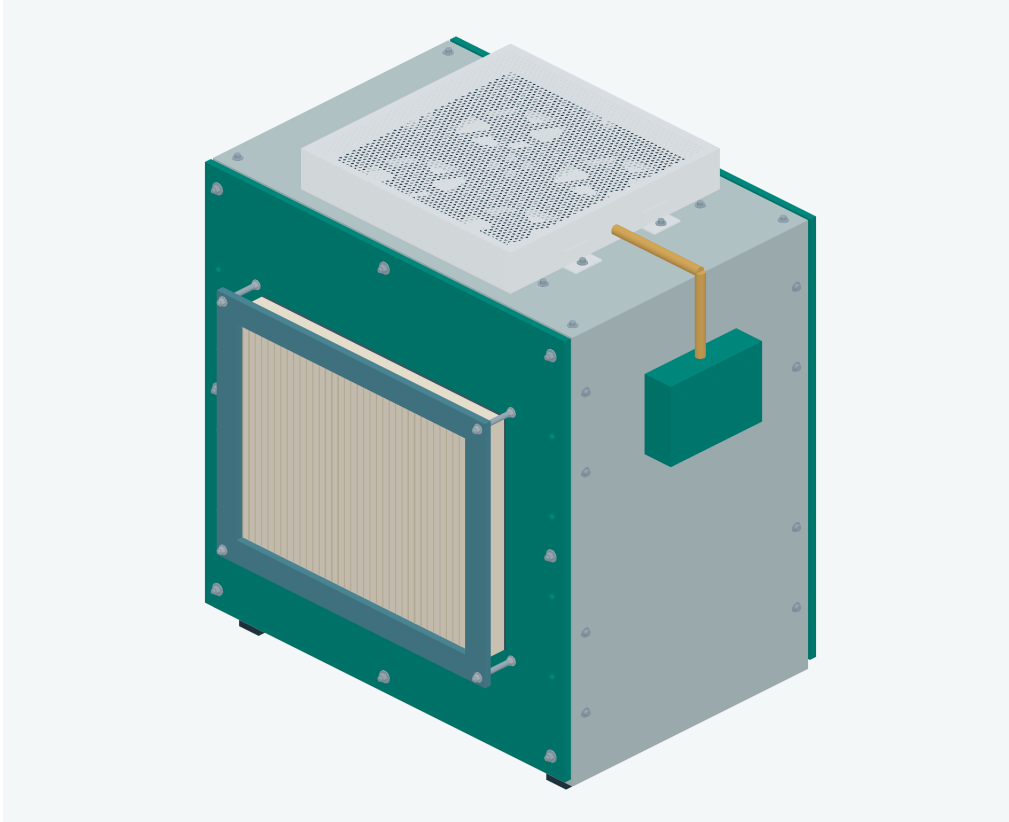


AQI Tower

Indoor prototype and next funding milestone

We are developing a fan-and-filter air cleaner. The first physical unit will help us measure how much air it moves, how well the filters work in our housing, and how practical it is to maintain.



Actual design rendering, not a photograph. The side box and yellow cable path are reserved space, not completed electrical hardware. Guard perforations are illustrative in the rendering; the native CAD faces remain solid envelopes.

Today: digital assembly and review documents exist. Physical prototypes built: **0**. This package asks for engineering review and staged funding, not permission to claim a finished product.

What the first unit will do

Air enters through two filters on opposite sides. Four fans draw that air into a shared chamber and discharge it through the top guard. The filters work side by side, not one after another.

Part	Plain explanation
Two removable filters	Catch particles; sealing around their edges is essential. Actual performance in this unit is untested.
Four fans	Move air through the resistance of filters and guards. Free-air fan figures are not the finished unit output.
Cabinet and guards	Hold the parts together and separate users from moving fans. Strength, access and electrical checks must finish before operation.
External power adapter	Proposed12V supply inside the fan system; household mains remains at the external OEM adapter. Electrical review is still required.

Why start smaller? It makes the first build easier to inspect, transport and measure. It remains our custom enclosure, filter mounting, integration and testing project. It does not prove that a large outdoor tower will work.

Long-term idea: larger towers may create local cleaner-air zones. A chain of zones is a research goal, not a proven network. Wind, weather, nearby pollution and placement will affect results. Bubble size, cleaning percentage and spacing are unknown.

No gas-removal, water, solar or outdoor coverage claim is attached to this first particulate prototype. The current experimental replacement filters do not make the assembled device a certified HEPA purifier.

What support achieves next

Already prepared	Still needed
Integrated CAD, exploded view, dimensioned panel drawings and hardware schedules	Internal mechanical review of structure, guarding, sealing and service
OEM fan data and pressure calculations	Real filter resistance and installed airflow measurements
Power-reference schematic and control requirements	Electrical design completion, rated protection and commissioning
Blank test sheets and checked analysis software	A supervised indoor location, instruments and an assembled unit

Funding request: fund engineering closure first, then released parts and outsourced fabrication, followed by qualified assembly and instrumented testing. Prices, stock and service lead times are not confirmed; the parts register is not a quotation.

Measurable next milestone: a reviewed revision with no unresolved safety-critical build holds, followed by one safely commissioned indoor unit and traceable airflow, pressure, power, noise and particle-test records.

Timing: internal review can start now. Cable drawings are requested from ARCTIC and Noctua. Supplier responses, workshop availability and rework govern the construction date; no completion date is booked.

Team: Nakul Kokate coordinates the project. Mechanical and electrical reviewers can work in parallel using the engineering pack. College and NGO support is funding; no college laboratory is assumed. A final project/device name remains to be agreed.